# Complications, outcomes, and implications of a prolonged vegetative state in anti-NMDA receptor encephalitis: a retrospective international cohort study

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## Summary

**Background** In anti-NMDA receptor (NMDAR) encephalitis, delayed recovery and slow improvement makes it difficult to assess treatment refractoriness, leaving a knowledge gap about prolonged impaired consciousness. We aimed to assess treatment response, long-term outcomes, and their predictors in patients with NMDAR encephalitis and a prolonged vegetative state.

**Methods** In this international retrospective cohort study, we included patients with NMDAR encephalitis from Jan 1, 2007 to Sept 30, 2024 who remained in a vegetative state (unresponsive wakefulness) for 9 months or longer in 21 hospitals across Austria, Brazil, Chile, China, Germany, Hong Kong, Japan, the Netherlands, Spain, Switzerland, and the USA. Data on disease presentation and long-term outcomes were collected via medical record review and a structured questionnaire sent to referring physicians. Outcomes were death, ability to follow commands, reaching a modified Rankin Scale (mRS) score of 2, or complete recovery (mRS 0 with return to premorbid activities). Competing risks analysis was used to assess survival and predictors of death, mRS score 2, and recovery.

**Findings** 45 patients were identified (38 female and seven male; median age 22 [IQR 19–31] years). The patients had impaired consciousness a median of 9 days after symptom onset (IQR 5–16). All 45 patients received first-line immunotherapy; 41 (91%) received second-line, and 18 (40%) received third-line immunotherapies. 21 (47%) of 45 patients had ovarian teratomas, which were removed in 20 patients. Median durations were: vegetative state 399 days (IQR 307–698); intensive care unit stay 275 days (189–354); mechanical ventilation 270 days (195–394); and hospitalisation 474 days (349–715). 13 (28%) patients were resuscitated from dysautonomic cardiac arrest. After a median of 5 years (IQR 3–7), 15 (33%) of 45 patients completely recovered (mRS score ≤1 and returned to all previous activities), 13 (29%) substantially improved (mRS score 2), 11 (24%) had mRS score 3–5, and six (13%) died. Five patients began command-following a median of 11 months (IQR 2–21) after last immunotherapy. Estimated cumulative incidence for reaching an mRS score of 2 was 66% at 5 years and 76% at 10 years, and for recovery was 32% at 5 years and 54% at 10 years. Teratomas were associated with a lower probability of reaching mRS score 2 (subdistribution hazard ratio 0·39 [95% CI 0·18–0·84], p=0·0160), whereas older age (1·10 per year [1·04–1·23], p=0·0052) and higher NMDAR encephalitis 1-Year Functional Status (NEOS)2 score (1·51 [1·12–2·04], p=0·0072) were associated with increased mortality.

**Interpretation** In people with NMDAR encephalitis and prolonged impairment of consciousness, full or substantial recovery was reached in approximately two-thirds of cases. Consequently, early assessment of treatment response or refractoriness might underestimate delayed improvement, prolonged therapy needs, or spontaneous recovery. Futility decisions should therefore be individualised with multidisciplinary input and based on extended follow-up.

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## Introduction

Anti-NMDA receptor (NMDAR) encephalitis is an autoimmune brain disorder characterised by CSF IgG antibodies against the GluN1 subunit of the NMDAR.[1] Three defining clinical features are the rapid onset of prominent psychosis or behavioural changes, often preceding or accompanying neurological symptoms; severe manifestations frequently requiring intensive

## Research in context

**Evidence before this study**

We searched PubMed for articles published in English from database inception until July 31, 2025, using the terms "anti-NMDA receptor encephalitis AND refractory" or "anti-NMDA receptor encephalitis AND treatment-resistant", and identified 50 articles (n=97 patients) predominantly reported as single case reports or small case series. Definitions and timelines of treatment refractoriness varied widely, with intervals between initiation of immunotherapy and determination of non-effectiveness ranging from 2 weeks to 39 months. Reported durations of vegetative or minimally conscious states ranged from 0 to 39 months; only 15 (16%) of 94 patients with assessable data remained in such a state for 9 months or longer. Among these 15 patients, five (33%) eventually improved, and the remaining ten (67%) developed persistent neurological deficits. This heterogeneity makes it unclear whether patients did not respond to treatment or experienced delayed recovery, complicating assessments of treatment efficacy and recovery potential in prolonged impaired consciousness. To our knowledge, no previous studies have specifically examined recovery potential in patients with NMDAR encephalitis with prolonged disorders of consciousness.

**Added value of this study**

This retrospective longitudinal study examines a unique cohort of 45 patients with NMDAR encephalitis and prolonged disorders of consciousness, characterised by a vegetative state (unresponsive wakefulness) lasting 9 months or longer. Despite prolonged ICU stays, mechanical ventilation, and frequent severe complications, nearly two-thirds of patients improved, including one-third who achieved complete recovery. Among those without improvement, approximately two-thirds had persistent moderate-to-severe disability and one-third died. Teratomas were associated with a lower probability of improvement, and older age and higher NMDAR encephalitis 1-Year Functional Status (NEOS)2 scores were associated with increased mortality.

**Implications of all the available evidence**

In NMDAR encephalitis, prolonged disorders of consciousness, such as vegetative and minimally conscious states, do not necessarily indicate irreversible brain injury. Assessments of treatment effects and refractoriness might overlook delayed responses, extended treatment needs, or spontaneous recoveries, all of which are important when evaluating novel or add-on therapies. Trials of third-line or experimental treatments should account for the complexity of measuring consciousness and treatment response in NMDAR encephalitis. The timing for declaring therapeutic futility remains undefined and should not be based on data from other conditions or chronic non-traumatic disorders of consciousness, as many patients with NMDAR encephalitis ultimately recover. Decisions about ongoing treatment and life support should be individualised, involving multidisciplinary teams, families, and ethical guidance.

care; and a delayed but typically gradual recovery following initiation of immunotherapy.[2]

Since its first report in 2007,[3] substantial progress has been made in understanding this disease, including its predominance in children and young adults, especially females;[4] the identification of an age-dependent symptom sequence (eg, children often initially show insomnia, seizures, or abnormal movements, and adolescents and adults more commonly present with insomnia and psychosis);[2] and the recognition of triggers such as ovarian teratoma or herpes simplex virus encephalitis.[3,5] Treatment approaches now include immunotherapy escalation and supportive care.[6] Antibodies are known to be pathogenic,[7] and clinical recovery depends on their removal, elimination of antibody-producing cells, and tumour resection when present.[2] Current treatments lead to recovery or substantial improvement in approximately 80% of patients. However, mortality remains at 5–8%, and the remaining patients who do not recover experience persistent, disabling deficits following minimal improvement or symptom stabilisation.[2] However, a small subset of patients deviates from this pattern, remaining in prolonged disorders of consciousness (vegetative or minimally conscious states) despite treatment, usually associated with abnormal movements, autonomic instability, or seizures.[8,9] As most patients with NMDAR encephalitis have impaired consciousness and treatment responses are typically delayed and slow, assessing treatment refractoriness is particularly challenging in prolonged unconscious states, raising complex clinical and ethical concerns.

In such cases, key clinical questions concern occurrence of in-hospital complications, the choice and duration of immunotherapy, long-term outcomes, and the point at which deficits might be considered irreversible and further treatment futile. We aimed to address these questions in a retrospective cohort of patients with NMDAR encephalitis who remained in a vegetative state or unresponsive wakefulness for 9 months or longer. We also assessed clinical predictors of outcome and treatment response.

## Methods

### Study design and participants

We conducted an international, retrospective cohort study of patients with idiopathic or tumour-related NMDAR encephalitis with prolonged disorders of consciousness, identified between Jan 1, 2007 and Sept 30, 2024 admitted for the acute stage of their disease in 21 hospitals across Austria, Brazil, Chile, China, Germany, Hong Kong, Japan, the Netherlands, Spain, Switzerland, and the USA. Inclusion criteria were patients of any age with 9 months or longer in a vegetative state, defined by absence of self-awareness or

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environment awareness; no sustained, reproducible, purposeful, or voluntary responses to visual, auditory, tactile, or noxious stimuli; no evidence of language comprehension or expression; and periodic eye opening[10,11] (with possible prolonged periods of eye closure). Exclusion criteria were patients with insufficient clinical data, and presence of multiple autoantibodies or NMDAR encephalitis features following herpes simplex encephalitis.

The threshold of 9 months or longer was arbitrarily chosen by the authors based on their personal experience with early patients (patients who were treated previously, close to the initial description of the disease, who improved or recovered after prolonged impaired consciousness).[8,12,13] Data on disease presentation and long-term outcomes were collected via medical record review and a structured questionnaire sent to referring physicians (appendix p 2). Symptom severity was assessed using the modified Rankin Scale (mRS) at disease peak, hospital discharge, and during follow-up (every 4 months), including the last visit (last available visit per routine clinical follow-up). At nadir (within the first month from hospital admission), we assessed the anti-NMDAR Encephalitis 1-Year Functional Status (NEOS) score (range 0–5)[14] and updated NEOS2 score (0–16),[15] with higher scores indicating lower 1-year probability of functional recovery, and the Clinical Assessment Scale in Autoimmune Encephalitis (CASE) score (0–27), with higher scores indicating greater autoimmune encephalitis severity.[16] Additional data included EEG, brain MRI, and CSF findings at onset and during follow-up. Functional status, residual symptoms, and patients' ability to resume their previous activities were recorded at the last follow-up visit. Four binary clinical outcomes and their dates of occurrence were evaluated: death; consistent ability to follow commands; reaching an mRS score of 2, defined as independence in managing personal affairs over 24 h despite reduced capacity for work, study, social interaction, or leisure; and complete recovery, defined as mRS score 0 and a full return to premorbid activities or employment without limitations or adaptations.

The study was approved by the Ethics Committee of Hospital Clínic of Barcelona (HCB/2018/0192). Written informed consent for the storage and research use of samples and clinical data was obtained from all participants or their legal proxies, as well as consent for publication from patients featured in the videos.

### Statistical analysis

Data are expressed as frequencies and percentages for categorical variables, and as median and IQR for continuous variables. Since the assumptions of normality were not met, non-parametric tests were applied. Comparisons of clinical features between NMDAR encephalitis subgroups categorised by outcomes were done with Chi-square test, Fisher's exact test, and Mann-Whitney U test, as appropriate. The (event-free) survival was evaluated with Fine-Gray competing risks survival analysis. We estimated the cumulative incidence functions from competing risks data across NMDAR encephalitis groups defined by two sets of outcomes: death or reaching mRS score 2, and death or reaching recovery. Univariable competing risks regression modelling was used to identify demographic, clinical, and paraclinical predictors of death and reaching mRS score 2. Sub-distribution hazard ratios (sHRs) and corresponding 95% CIs were estimated for each predictor. We compared first-line and second-line immunotherapies versus combined with third-line (add-on) immunotherapies on the same two competing outcomes; sHRs were calculated to estimate the relative effect of treatment escalation on each outcome, adjusting for sex, age, and time from onset to immunotherapy initiation as potential confounders. Survival analysis was performed with the library cmprsk (2.2–12). The level of significance was established at the two-sided 5%, using IBM SPSS Statistics (version 26) and R statistical computing software for Windows (version 4.4.2).

### Role of the funding source

The funder of the study had no role in the study design, data collection, data analyses, data interpretation, or writing of the report.

## Results

We identified 47 patients with NMDAR encephalitis who fulfilled the inclusion criteria; two of whom were excluded due to insufficient information. Nine patients were previously reported (seven by TI or JD), but with shorter follow-up or emphasis on clinical-radiological features rather than treatment refractoriness or vegetative state (appendix p 3). In 28 (62%) of 45 patients, diagnosis and clinical course were documented while the patients were in a vegetative state through ongoing consultations between the treating neurologists and authors (TI and JD); for the remainder, data were obtained during study planning with collaborating investigators.

The median age at symptom onset was 22 years (IQR 19–31; range 2–43), including two patients younger than 12 years. 38 (84%) patients self-identified as female and seven (16%) as male. Demographic and clinical features at onset are summarised in table 1 and the appendix (p 3). Two patients had a previous NMDAR encephalitis episode, with good recovery 6 and 9 years earlier (appendix pp 3–4). All patients developed impaired consciousness a median of 9 days (IQR 5–16) after symptom onset and all were admitted to the intensive care unit (ICU). Further indications for ICU admission are indicated in the appendix (p 4).

Initial EEG, MRI, and CSF studies were obtained in all patients within the first month of symptom onset. At presentation, all patients had EEG alterations, with extreme delta brush observed in 12 (27%) initially and in 25 (56%) within the first month; MRI changes occurred in 19 (42%), and CSF pleocytosis in 43 (98%) of 44 (table 1 and appendix pp 18–19).

All patients received first-line immunotherapy (intravenous steroids, immunoglobulins, and/or plasma exchange) initiated a median of 12 days (IQR 8–20) from onset. Second-line immunotherapy (rituximab or cyclophosphamide, or both, with or without steroid-sparing agents such as azathioprine or mycophenolate mofetil) was given to 41 patients (91%) at a median of 36 days (IQR 27–71) from onset, and third-line immunotherapy was administered to 18 patients (40%) at a median of 193 days (IQR 121–263). These included one or more of the following: bortezomib in 16 patients (36%); methotrexate in seven (16%); tocilizumab in three (7%); daratumumab in two (4%); and another immunotherapy in seven (16%; table 1).

For each immunotherapy, the number of courses, initiation relative to symptom onset, and timing of oral steroid tapering are shown in table 2. The three main reasons for changing or discontinuing immunotherapy were infections (11, 24%), reaching cumulative dose limits or adverse events (eight, 18%), and the perception of treatment inefficacy (five, 11%). The most frequently used symptomatic treatments are reported in the appendix (pp 4–5).

Tumours were identified in 21 (47%) of 45 patients, all were ovarian teratomas, including six bilateral. Surgery was performed in 20 patients (median, 24 days from symptom onset, IQR 14–44); one patient was not operated due to clinical instability. Of the 26 teratomas removed, 17 were mature cystic, four immature grade 1, and five unspecified. For additional information and empiric oophorectomies see the appendix (p 5).

Patients remained in a vegetative state for a median of 399 days (IQR 307–698), with a median ICU duration of 275 days (189–354) and a mechanical ventilatory support duration of 270 days (195–394). Median hospitalisation was 474 days (IQR 349–715); 28 (64%) patients were transferred to rehabilitation or long-term care, nine (20%) were discharged home, six (13%) died, and two (4%) remained hospitalised at last follow-up (table 1).

During hospitalisation, 44 (98%) patients underwent percutaneous endoscopic gastrostomy, and 43 (96%) had a tracheostomy. Most relevant ICU complications were recurrent infections in 41 (91%) patients and thrombotic events in 26 (58%), including pulmonary embolism in 13 (29%). 13 (28%) patients were resuscitated from dysautonomic cardiac arrest or asystole lasting ≥30 s, at a median of 28 days (IQR 23–146) from disease onset. Five underwent cardiopulmonary resuscitation twice (median interval 6 days), two required it three times, and two needed a temporary external pacemaker. All cardiac arrests occurred in the ICU, without evidence of ischaemic brain injury: among the 13 patients who were resuscitated, three (23%) eventually reached full recovery, three (23%) reached mRS score 2, seven (54%) had an

| | All patients (n=45) | mRS 0–2 (n=28) | mRS 3–6 (n=17) | p value for mRS 0–2 *vs* mRS 3–6 |
|---|---|---|---|---|
| Age, years | 22 (19–31) | 22 (21–31) | 25 (18–31) | 0·87 |
| Sex | .. | .. | .. | 0·39 |
| Female | 38 (84%) | 25 (89%) | 13 (76%) | .. |
| Male | 7 (16%) | 3 (11%) | 4 (24%) | .. |
| First main clinical features | .. | .. | .. | 0·84 |
| Behavioural or psychiatric | 23 (51%) | 14 (50%) | 9 (53%) | .. |
| Cognition | 7 (16%) | 5 (18%) | 2 (12%) | .. |
| Seizure | 11 (24%) | 6 (21%) | 5 (29%) | .. |
| Other | 4 (9%) | 3 (11%) | 1 (6%) | .. |
| Time from onset to impaired consciousness, days | 9 (5–16) | 9 (6–22) | 8 (3–12) | 0·24 |
| EEG at presentation* | | | | |
| Unremarkable | 0 | 0 | 0 | .. |
| With extreme delta brush | 12 (27%) | 7 (25%) | 5 (29%) | 0·50 |
| Other changes | 33 (73%) | 21 (75%) | 12 (71%) | 0·75 |
| EEG within the first month, with extreme delta brush | 25 (56%) | 12 (43%) | 13 (76%) | 0·0284 |
| Brain MRI at presentation* | | | | |
| Unremarkable | 26 (58%) | 18 (64%) | 8 (47%) | 0·26 |
| With T2/FLAIR hyperintensities | 16 (36%) | 9 (32%) | 7 (41%) | 0·54 |
| Other changes | 3 (7%) | 1 (4%) | 2 (12%) | 0·32 |
| CSF studies at diagnosis* | | | | |
| Unremarkable | 1/44 (2%) | 1/27 (4%) | 0/17 | 0·99 |
| Pleocytosis | 43/44 (98%) | 26/27 (96%) | 17/17 (100%) | 0·99 |
| WBC, μL | 65 (28–150) | 56 (25–100) | 88 (29–203) | 0·31 |
| Tumour | 21 (47%) | 10 (36%) | 11 (65%) | 0·0591 |
| Bilateral ovarian teratoma | 6/21 (29%) | 2/10 (20%) | 4/11 (36%) | 0·37 |
| mRS at diagnosis | 5 (3–5) | 5 (4–5) | 4 (3–5) | 0·30 |
| CASE score at diagnosis | 26 (24–26) | 25 (24–26) | 26 (25–27) | 0·11 |
| NEOS score | 3 (3–4) | 3 (3–4) | 4 (3–4) | 0·18 |
| NEOS2 score | 8 (6–8) | 7 (6–8) | 8 (7–8) | 0·38 |
| First-line immunotherapies | 45 (100%) | 28 (100%) | 17 (100%) | .. |
| Days after symptom onset | 12 (8–20) | 13 (8–20) | 10 (7–15) | 0·20 |
| Corticosteroids† | 45 (100%) | 28 (100%) | 17 (100%) | .. |
| Intravenous immunoglobulins | 37 (82%) | 23 (82%) | 14 (82%) | 0·99 |
| Plasma exchange | 34 (76%) | 22 (79%) | 12 (71%) | 0·72 |
| Second-line immunotherapies‡ | 41 (91%) | 26 (93%) | 15 (88%) | 0·63 |
| Days after symptom onset | 36 (27–71) | 37 (29–86) | 32 (24–45) | 0·23 |
| Rituximab§ | 34 (76%) | 22 (79%) | 12 (71%) | 0·72 |
| Cyclophosphamide | 32 (71%) | 20 (71%) | 12 (71%) | 0·99 |
| Third-line immunotherapies | 18 (40%) | 8 (29%) | 10 (59%) | 0·0451 |
| Days after symptom onset | 193 (121–263) | 178 (119–413) | 193 (121–263) | 0·99 |
| Bortezomib | 16 (36%) | 8 (29%) | 8 (47%) | 0·21 |
| Methotrexate¶ | 7 (16%) | 3 (11%) | 4 (24%) | 0·40 |
| Tocilizumab | 3 (7%) | 0 | 3 (18%) | 0·0480 |
| Daratumumab | 2 (4%) | 2 (7%) | 0 | 0·52 |
| Other immunotherapies‖ | 7 (16%) | 2 (7%) | 5 (29%) | 0·061 |
| Duration of vegetative state, days | 399 (307–698) | 366 (298–536) | 650 (399–1031)** | 0·0042 |
| Duration of ICU admission, days | 275 (189–354) | 264 (184–343) | 270 (203–360) | 0·81 |
| Duration of OTI and ventilatory support, days | 270 (195–394) | 238 (177–355) | 305 (228–470) | 0·13 |

(Table 1 continues on next page)

| | All patients (n=45) | mRS 0–2 (n=28) | mRS 3–6 (n=17) | p value for mRS 0–2 vs mRS 3–6 |
|---|---|---|---|---|
| (Continued from previous page) | | | | |
| Duration of hospital admission, days | 474 (349–715) | 442 (338–574) | 524 (417–779) | 0·12 |
| Duration of total follow-up, years | 5 (3–7) | 6 (4–8) | 2 (2–5) | <0·0001 |
| EEG at last follow-up test available* | | | | |
| Unremarkable | 9/42 (21%) | 8/26 (31%) | 1/16 (6%) | 0·060 |
| With alterations | 33/42 (79%) | 18/26 (69%) | 15/16 (94%) | .. |
| Absent posterior α rhythm | 24/42 (57%) | 10/26 (38%) | 14/16 (88%) | 0·0025 |
| Brain MRI at last follow-up test available* | | | | |
| Unremarkable | 17/43 (40%) | 13/26 (50%) | 4/17 (24%) | 0·093 |
| Residual global atrophy | 23/43 (53%) | 10/26 (38%) | 13/17 (76%) | 0·0154 |
| New white matter lesions | 3/43 (7%) | 2/26 (8%) | 1/17 (6%) | 0·66 |
| CSF studies at last follow-up sample available* | | | | |
| Unremarkable | 14/42 (33%) | 12/25 (48%) | 2/17 (12%) | 0·0163 |
| Pleocytosis | 12/42 (29%) | 6/25 (24%) | 6/17 (35%) | 0·43 |
| Functional status along disease course | | | | |
| mRS at 1-year follow-up | 5 (4–5) | 4 (4–5) | 5 (5–5) | 0·0001 |
| mRS at hospital discharge | 4 (3–4) | 4 (3–4) | 5 (4–5) | 0·0001 |
| Functional status at last follow-up | | | | |
| mRS 0 | 7 (16%) | .. | .. | .. |
| mRS 1 | 8 (18%) | .. | .. | .. |
| mRS 2 | 13 (29%) | .. | .. | .. |
| mRS 3 | 1 (2%) | .. | .. | .. |
| mRS 4 | 3 (7%) | .. | .. | .. |
| mRS 5 | 7 (16%) | .. | .. | .. |
| mRS 6 (death) | 6 (13%) | .. | .. | .. |

Data are shown as n (%) or median (IQR). Sex was self-reported. CASE=clinical assessment scale in autoimmune encephalitis. ICU=intensive care unit. mRS=modified Rankin scale. NEOS=anti-NMDAR encephalitis 1-year functional status score. OTI=orotracheal intubation. T2/FLAIR=T2-weighted/fluid attenuated inversion recovery sequences. WBC=white blood cell count. *Additional information in the appendix (p 6, 18–19). †Intravenous pulses of methylprednisolone in all cases with or without being followed by tapering oral steroids. ‡Additional use of azathioprine in seven (16%) patients and mycophenolate mofetil in four (9%) patients. §Additional doses intrathecally administered in two patients. ¶Intrathecal administration in three patients. ||Added in combination to the previous immunotherapies: efgartigimod alfa (anti-neonatal Fc receptor) in two (4%); CSF apheresis in two (4%); tacrolimus in one (2%); alemtuzumab in one (2%); and haematopoietic stem-cell transplantation in one (2%). **Four patients stayed in a vegetative state for over 4 years (median 5·2 years; IQR 4·4–10·4), despite receiving immunotherapy within 10 days of symptom onset and treatment with multiple second-line and third-line agents.

***Table 1:*** **Clinical features, immunotherapy, and outcomes stratified by mRS***

See Online for videos 1–3

mRS score of 3 or greater, and three later died of other complications.

In four patients, intractable abnormal movements caused bone fractures or dental trauma. Other complications during ICU stay are listed in the appendix (pp 5–6).

Follow-up EEG, MRI, and CSF findings are presented in table 1 and the appendix (p 6, 18–19). Follow-up of CSF antibody titres was available from 30 patients (median 505 days post-onset; IQR 349–782); all remained NMDAR antibody-positive, with high titres (≥1:64) in 23 (77%). High titres showed no association with outcome, occurring in 14 (78%) of 18 patients with an mRS score of 0–2 and nine (75%) of 12 with an mRS score of 3–6 (p=0·60).

After a median follow-up of 5 years (IQR 3–7), 15 (33%) of 45 patients fully recovered (mRS ≤1 and returned to all previous activities), 13 (29%) of 45 substantially improved (mRS 2) including the two children, 11 (24%) of 45 remained with an mRS score of 3–5, and six (13%) of 45 had died (table 3). The timings of death and of improvement milestones (ability to follow commands, mRS score 2, or full recovery) measured from symptom onset and from initiation of immunotherapies are provided in table 4.

Among the 28 patients who improved, the duration of the vegetative state ranged between 277 and 1405 days, whereas in the 11 who remained with an mRS score 3–5, it persisted for 361–3814 days (minimum follow-up 557 days; seven were followed-up for more than 1405 days). Six patients never responded to commands, four had the ability to follow commands but without further progress, and one reached an mRS score of 3. The main causes of death were infections (three [50%] of six) and withdrawal of care (two [33%] of six). Regarding treatment, eight (29%) of 28 patients who improved received third-line immunotherapies, compared with ten (59%) of 17 who did not improve or died (p=0·045).

Five patients remained in a vegetative state for a median of 24·5 months (IQR 23–37; range 21–47) and had the longest intervals between their last immunotherapy and improvement milestones: command-following (median 11 months, IQR 2–21) and mRS score 2 (median 22 months, IQR 18–48). Despite these prolonged delays, at a median follow-up of 11 years (IQR 6–14), three patients had fully recovered (mRS score 0, return to all previous activities), and the remaining two exhibited only mild cognitive deficits and remained socially active and driving (appendix p 7).

One patient recovered despite a 9-month delay in receiving immunotherapy and undergoing ovarian teratoma removal (appendix pp 7–9; p 22; video 1). Three patients received first-line immunotherapy for 5–13 months without meaningful improvement, before responding to second-line treatment.

One patient showed substantial improvement (mRS score 1) despite a protracted course with multiple complications (appendix pp 9–12; p 23; video 2). One patient had no meaningful improvement after 4 years of intensive therapy, including haematopoietic stem-cell transplantation (appendix pp 12–17; p 24; video 3).

All deaths occurred between 10 and 28 months after disease onset (figure), with mortality gradually rising over the first 2 years and plateauing at approximately 13% by 2·5 years. Among survivors, the estimated cumulative incidence from competing risks analysis of reaching mRS score 2 steadily increased, accelerating within the first 3 years. By 5 and 10 years, 66% and 76% of the cohort, respectively, had reached mRS score 2 (figure). The estimated cumulative incidences of complete recovery were 32% by 5 years and 54% by 10 years (appendix p 25).

| | All patients (n=45) | mRS 0–2 (n=28) | mRS 3–6 (n=17) | p value for mRS 0–2 *vs* mRS 3–6 |
|---|---|---|---|---|
| **First-line immunotherapies, number of courses; days after symptom onset** | | | | |
| Corticosteroids, intravenous pulses | 3 (2–6); 12 (8–19) | 3 (2–6); 10 (8–15) | 3 (2–6); 14 (8–20) | 0·88; 0·21 |
| Intravenous immunoglobulins | 3 (1–5); 15 (9–42) | 3 (1–5); 15 (9–45) | 4 (1–7); 14 (10–30) | 0·36; 0·58 |
| Plasma exchange | 5 (2–7); 20 (15–39) | 5 (2–10); 27 (16–37) | 3 (2–6); 18 (13–40) | 0·43; 0·22 |
| Corticosteroids, oral taper days of duration | 90 (59–185) | 87 (64–395) | 105 (63–150) | 0·92 |
| **Second-line immunotherapies, number of courses; days after symptom onset** | | | | |
| Rituximab | 4 (2–5); 33 (25–52) | 4 (2–4); 33 (28–55) | 6 (4–7); 33 (25–46) | 0·026; 0·61 |
| Cyclophosphamide | 6 (5–8); 58 (37–180) | 6 (6–8); 58 (41–101) | 6 (5–7); 66 (32–244) | 0·56; 0·97 |
| **Third-line immunotherapies*, number of courses; days after symptom onset** | | | | |
| Bortezomib | 4 (3–6); 207 (122–471) | 5 (4–6); 178 (119–413) | 4 (4–7); 211 (157–402) | 0·99; 0·60 |
| Methotrexate | 5 (2–15); 223 (131–429) | 2 (1–15); 213 (200–343) | 4 (2–15); 335 (186–671) | 0·72; 0·48 |
| Tocilizumab | 3 (2–6); 442 (180–792) | 0 | 3 (2–6); 442 (180–792) | .. |
| Daratumumab | 15 (10–19); 248 (196–300) | 15 (10–19); 248 (196–300) | 0 | .. |

Data are median (IQR). *All 18 patients (40%) who received third-line treatments had also been treated with second-line immunotherapies.

***Table 2:*** **Number of courses and initiation relative to symptom onset of immunotherapy and timing of steroid tapering**

| | First main clinical feature | Brain MRI | EEG | CSF studies | Tumour | Immunotherapies* | From symptom onset to impaired consciousness, and death† | Medical complications and cause of death |
|---|---|---|---|---|---|---|---|---|
| Male, 38 years[17] | Seizures | Onset and at 1·5 months: unremarkable; at 4·8 months and 8·8 months: progressive diffuse brain atrophy | Onset: unremarkable; day 10: diffuse delta slowing; at 9·9 months: diffuse delta slowing | Day 4: 279 WBC/μL, proteins 149 mg/dL; day 182: 0 WBC/μL, proteins 25 mg/dL, IgG index 0·85, OCB (+) | No | First-line: 4 cycles of IVIg from day 43; 3 rounds of PLEX (5 sessions) from day 50; 3 cycles of IVMP from day 161, followed by oral prednisolone 30 mg/day | Unresponsive from day 4; died 10·3 months after onset | Initially suspected HHV-6 infection (later found to be chromosomal integration instead of CNS infection); died of septic shock, acute kidney injury and cardiopulmonary collapse |
| Female, 31 years | Fever and behavioural alterations | Onset: T2/FLAIR hyperintensity in medial temporal lobes, mild left frontal meningeal enhancement; at 3 months: diffuse brain atrophy | Onset: EDB | Onset: 108 WBC/μL, proteins 52 mg/dL, OCB (+); day 14: 30 WBC/μL, proteins 31 mg/dL | Left ovarian teratoma removed on day 4, right oophorectomy (no tumour)‡ | First-line and second-line: 2 cycles of IVMP from day 3, 1 cycle of IVIg from day 5, 5 cycles of CTX from day 28, chronic oral prednisolone and AZA | Impaired consciousness from day 3; died 12 months after onset | Gastrointestinal bleeding and sepsis |
| Male, 37 years | Behavioural alterations | Onset: unremarkable; at 14·3 months: diffuse brain atrophy | At 3 and 12 months: EDB | Onset: 20 WBC/μL, proteins 35 mg/dL | No | First-line, second-line, and third-line: 2 cycles of IVMP from day 18, 3 cycles of IVIg from day 30, 4 doses of RTX from day 45, 3 cycles of BTZ from day 96, 2 doses of efgartigimod from day 365 | Impaired consciousness from day 16; died 14 months after onset | Malignant hyperthermia, venous thromboembolism, multiple infections, oesophagotracheal fistula that required surgery; died of septic shock |

(Table 3 continues on next page)

Among all variables assessed, the presence of an ovarian teratoma was significantly associated with a lower probability of reaching an mRS score of 2 (sHR 0·39 [95% CI 0·18–0·84], p=0·0160), whereas older age at onset (1·10 per year [1·04–1·23], p=0·0052) and higher NEOS2 score during the acute phase of disease (1·51 [1·12–2·04], p=0·0072) were the only significant predictors of death. No other demographic, clinical, or paraclinical variables predicted recovery or were associated with mortality (all p>0·05; appendix p 20).

Patients treated with only first-line and second-line therapies (ie, steroids, immunoglobulins, plasma exchange, rituximab, and cyclophosphamide) showed a trend toward a higher probability of reaching an mRS score of 2 compared with those who also received third-line immunotherapies (sHR 0·40 [95% CI 0·15–1·08], p=0·06; appendix p 21).

| | First main clinical feature | Brain MRI | EEG | CSF studies | Tumour | Immunotherapies* | From symptom onset to impaired consciousness, and death† | Medical complications and cause of death |
|---|---|---|---|---|---|---|---|---|
| (Continued from previous page) | | | | | | | | |
| Female, 33 years | Behavioural alterations | Onset and at 17 months: unremarkable | Onset: status epilepticus with left temporal onset and secondary generalisation; at 34 months: slow background activity | Onset: 230 WBC/μL, proteins 35 mg/dL, OCB (+); at 17·2 months: 1 WBC/μL, proteins 20 mg/dL, OCB (+) | Left ovarian teratoma removed on day 20, right oophorectomy (no tumour) | First-line, second-line, and third-line: 1 cycle of IVMP from day 2, 3 cycles of IVIg from day 11, 2 rounds of PLEX (7 sessions) from day 18, 2 doses of RTX from day 32, 2 cycles of IT MTX/prednisolone from day 131, 6 cycles of CTX from day 212, 3 cycles of BTZ from day 212 | Impaired consciousness from day 1; died 21 months after onset | Recurrent infections, venous thrombosis, severe bradycardia, ventilator-dependent; withdrawal of care |
| Female, 39 years[9] | Behavioural alterations | Onset: diffuse leptomeningeal enhancement; at 4 months: unremarkable; at 24 months: diffuse brain atrophy | Onset: diffuse delta slowing | Onset: 237 WBC/μL, proteins 65 mg/dL; at 16 months: unremarkable | Right ovarian teratoma removed on day 50, left oophorectomy (no tumour) | First-line, second-line, and third-line: 4 cycles of IVIg from day 52, 1 cycle of IVMP from day 56, 6 doses of RTX from day 70, 6 cycles of CTX from day 183, 1 round of PLEX (6 sessions) from month 12, 5 cycles of IV high-dose MTX from month 13 | Impaired consciousness from day 9; died 25 months after onset | Multiple systemic complications and sepsis; withdrawal of care |
| Female, 21 years | Psychosis | Onset: T2/FLAIR hyperintensities in hippocampi and basal ganglia; at 6 months: resolution of hyperintensities, new contrast enhancement and meningeal thickening with frontotemporal atrophy; at 11 months: global atrophy | Onset: severely disorganised and slow background activity; day 15: similar findings and epileptiform activity; at 6 months: EDB and epileptiform activity | Onset: 71 WBC/μL, proteins 333 mg/dL, OCB (+); at 10 months: 1 WBC/μL, proteins 973 mg/dL, IgG index 0·54, OCB (+); at 14 months: 0 WBC/μL, proteins 867 mg/dL, IgG index 0·48, OCB (-) | Bilateral ovarian teratomas (right removed on day 17; left on day 184) | First-line, second-line, and third-line: 3 rounds of PLEX (5 sessions) from day 7, 2 doses of RTX from day 27, 2 cycles of IVMP from day 37, 6 cycles of BTZ from day 121, 4 cycles of CTX from day 328, 1 cycle of alemtuzumab and IT MTX from 14 months | Impaired consciousness from day 5; died 27 months after onset | Dysautonomia, multiple infections or sepsis, thromboses, pulmonary embolism, malignant hyperthermia, abdominal compartment syndrome, acute kidney failure (haemodialysis), bone fractures, pneumothorax, resuscitated from cardiac arrest twice |

AZA=azathioprine. BTZ=bortezomib. CTX=cyclophosphamide. EDB=extreme delta brush. HHV-6=human herpesvirus 6. IT=intrathecal. IV=intravenous. IVIg=intravenous immunoglobulins. IVMP=intravenous methylprednisolone. MTX=methotrexate. OCB=oligoclonal bands. PLEX=plasma exchange. RTX=rituximab. T2/FLAIR=T2-weighted/fluid-attenuated inversion recovery sequences. WBC=white blood cells. *All patients (six) received first-line immunotherapy; five received second-line, and four received third-line. †All patients died while in vegetative or minimally conscious states. ‡History of removal of bilateral ovarian teratomas 5 years earlier.

*Table 3:* **Demographic and clinical characteristics of patients who died**

## Discussion

This study shows that NMDAR encephalitis can cause exceptionally prolonged disturbances of consciousness, akin to a vegetative state, yet still allow for meaningful delayed recovery. After at least 9 months in this state and a median follow-up of nearly 5 years (IQR 3–7 years), 62% of patients improved to an mRS score of 2 or less, including one-third who reached complete recovery. Among the remaining 38% of patients, approximately two-thirds had persistent moderate-to-severe disability, with most unable to follow commands, and the other third died. Most deaths occurred during the first 2 years, whereas functional recovery often continued gradually over several years.

The unexpectedly high recovery rate is remarkable, given that vegetative states are typically considered chronic—and rarely reversible—after 1 year in traumatic cases or 6 months in non-traumatic ones (usually post-anoxic, metabolic, or degenerative).[10,18] NMDAR encephalitis was not considered when current classification guidelines for prolonged disorders of consciousness were established.[18–20] These recoveries suggest that NMDAR encephalitis involves primarily functional, not structural, brain injury—unlike the conditions that shaped earlier definitions.

The rapid decline in consciousness within 2 weeks, CSF pleocytosis in 98% of patients, and EEG abnormalities in all (over 50% showing extreme delta

| | N of patients (N=45) | Time from onset, days | Time from initial first-line immunotherapy, days | Time from initial second-line immunotherapy, days | Time from initial third-line immunotherapy, days | Time from last immunotherapy,* days |
|---|---|---|---|---|---|---|
| Death† | 6 (13%) | 530 (348–805) | 520 (330–755); n=6 | 607 (355–754); n=5 | 508 (324–678); n=4 | 149 (35 to 288) |
| Follow commands† | 33 (73%) | 391 (298–525) | 367 (286–517); n=33 | 342 (223–407); n=31 | 206 (144–386); n=9 | 22 (–80 to 254) |
| mRS 2† | 28 (62%) | 753 (580–1329) | 722 (536–1268); n=28 | 674 (493–1159); n=26 | 587 (436–762); n=8 | 424 (124 to 755) |
| Complete recovery (return to premorbid activities)† | 15 (33%) | 1394 (864–1882) | 1382 (852–1843); n=15 | 858 (836–1364); n=13 | 919 (389–1306); n=4 | 797 (577 to 1316) |

Data are shown as n (%) or median (IQR). *Defined as the final dose of any second-line or third-line immunotherapy administered, or, for patients who did not receive second-line or third-line therapies, the final dose of intravenous first-line immunotherapy. †Refers to the time to death, or the time of achieving command-following, a functional status consistent with mRS 2, or complete recovery.

*Table 4:* **Timings of death and milestones of improvement from symptom onset and initiation of immunotherapies**

brush *vs* <30% in most series[21–23]) reflect disease severity. ICU admission was usually prompted by hallmark features of the disease, including autonomic instability, hypoventilation, seizures, intractable movement disorders, or severe agitation. Complications were frequent and primarily infections and thrombotic or haemorrhagic events. Notably, 28% of patients required resuscitation from dysautonomic cardiac arrest, four-times higher than in a previous ICU-focused NMDAR encephalitis study, highlighting the exceptional severity of this cohort.[24] Despite this level of severity, no patients died from cardiac arrest, and nearly half of patients who were resuscitated from cardiac arrests reached an mRS score of 0–2.

Consistent with previous NMDAR encephalitis reports, where ovarian teratoma detection and resection were not linked to better outcomes compared with non-tumour cases,[2] or were more common in refractory cases,[25] we found the presence of teratoma to be significantly associated with a lower probability of reaching an mRS score of 2. This finding suggests a more aggressive disease course in paraneoplastic cases, despite early immunotherapy and tumour removal. Older age at onset and a higher NEOS2 score emerged as the only significant predictors of mortality. Although not significant, early MRI alterations and status epilepticus showed trends toward worse outcomes, warranting further investigation.

Patients treated only with first-line and second-line immunotherapies showed a non-significant trend toward better functional outcomes than those receiving third-line therapies. Although this might suggest that third-line therapies do not improve recovery or reduce mortality, the finding more likely reflects confounding by indication, as escalation to third-line therapies is often driven by greater disease severity, treatment resistance, or anticipated poor prognosis.

In the largest observational NMDAR encephalitis study to date (n=577), published over a decade ago, a stepwise approach—first-line therapy (steroids, immunoglobulins, or plasma exchange), followed by second-line agents (rituximab or cyclophosphamide) as needed—resulted in

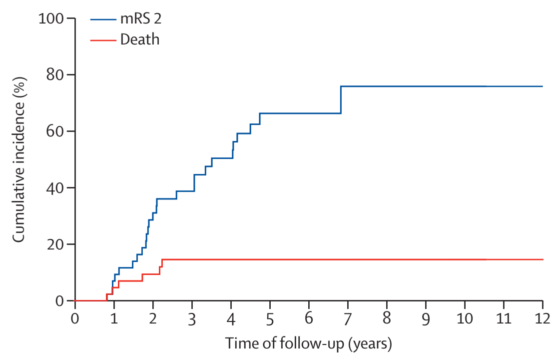


*Figure:* **Competing risks analysis of reaching mRS 2 versus death**
Competing risks analysis evaluating two mutually exclusive clinical outcomes: reaching a favourable functional status (mRS 2; n=28 patients; blue) and death (n=6 patients; red). Cumulative incidence curves are plotted over a 10-year follow-up. The incidence of mRS 2 increases steadily, reaching approximately 76% by year 7, and mortality plateaus at approximately 15% after 2·5 years. mRS=modified Rankin scale.

approximately 80% of patients reaching mRS score 0–2 at 2 years.[2] No alternative strategy has since shown better outcomes. Cyclophosphamide, a recognised second-line treatment, is used less frequently than other treatments, with increasing preference for agents such as bortezomib, tocilizumab, and others, despite scarce evidence, potential reporting bias, and confounding by previous or concurrent second-line therapies.[26–28] However, given the frequent use and efficacy of second-line therapies, assessing their earlier implementation (eg, as part of first-line therapy) might be warranted, alongside consideration of third-line options.

Spontaneous recovery has been reported in NMDAR encephalitis.[2,8] In our cohort, one patient recovered from a previous NMDAR encephalitis episode without immunotherapy. In five others, meaningful milestones were reached at median intervals of nearly 1 year (command-following) and 2 years (mRS 2) after their last treatment. These prolonged gaps raise the possibility of spontaneous improvement after extended periods in a vegetative state, although a delayed effect of immunotherapy cannot be excluded. Historical experience, drawn from times of limited disease recognition,[8] retrospective postmortem diagnoses,[3] or treatment before widespread use of second-line therapies,[2] suggests such spontaneous recoveries are exceedingly rare. Appendix pp 7–9 describes a patient who remained in a vegetative state for 9 months due to a missed diagnosis and lack of targeted treatment but ultimately made a full recovery after NMDAR encephalitis was identified and treated. This case, unlikely to represent spontaneous recovery, demonstrates that meaningful improvement is possible even after prolonged delays in treatment.

Slow response to immunotherapy and prolonged recovery are well-documented in NMDAR encephalitis,[1,2] yet this understanding remains insufficiently integrated into clinical practice. Definitions of refractoriness—often prompting changes or additions to therapy—vary widely, from 2 weeks to 39 months. Over half of studies using terms like “refractory” or “super-refractory” NMDAR encephalitis define refractoriness within 3 months of treatment initiation, a timeframe that might be too short for this disorder. We used a 9-month vegetative state as an extreme model to examine recovery potential in patients who do not respond early to immunotherapy or remain untreated. Our findings support the possibility of late improvement and challenge interpretations of treatment efficacy outside clinical trials or without a clear definition of refractoriness in NMDAR encephalitis, suggesting that outcomes credited to third-line or unconventional therapies might instead reflect delayed effects of second-line treatment, or rarely, spontaneous recovery.

This study has several considerations and limitations. Terminology for prolonged disorders of consciousness remains imprecise; terms such as vegetative state and unresponsive wakefulness refer to the same condition but both have limitations.[10,11,18] Based on current guidelines, vegetative state was the most appropriate classification for our patients. Although most required periods of sedation, ICU and neurology teams regularly assessed awareness and command-following during light or no sedation. Following sedation cessation after improvement in seizures, dyskinesias, agitation, or dysautonomia, patients remained in a vegetative state for an overall duration of at least 9 months from disease onset. During this period, none showed reproducible signs of awareness (eg, consistent eye tracking and command-following) that define the minimally conscious state, although those who improved passed through this state in recovery. Brief episodes of coma (absence of wakefulness and awareness) might have occurred, but the prolonged condition was most consistent with a vegetative state.

Other limitations of this study are the observational design and treatment heterogeneity, particularly beyond conventional first-line and second-line therapies. The exceptional nature of the cohort, assembled over many years, also precluded standardised biomarker evaluation beyond antibody testing. CSF antibodies are more reflective of disease activity than serum. All assessable patients had detectable antibodies a median of 505 days (IQR 349–782) after onset, with 75% at high titres. MRI and EEG results were provided by treating physicians and were not centrally reviewed. Nonetheless, despite non-standardised MRI follow-up, our observations are consistent with previous reports indicating that brain atrophy might be reversible in some patients, cautioning against its interpretation as a definitive marker of poor prognosis.[13,29]

These findings have important implications for clinical practice and disease understanding. First, prolonged disorders of consciousness, like vegetative and minimally conscious states, should not be equated with irreversible brain injury in NMDAR encephalitis. As with the distinction between traumatic and non-traumatic causes, within non-traumatic cases the underlying cause (autoimmune *vs* post-anoxic, metabolic, or degenerative[10,18]) is more predictive of outcome than the duration of unconsciousness. Second, current definitions of treatment refractoriness might underestimate the necessary duration of therapy in NMDAR encephalitis. Third, the timing for declaring therapeutic futility remains undefined and should not rely on experience from other conditions or recovery estimates for other causes of chronic non-traumatic vegetative state,[11,18] as many patients with NMDAR encephalitis eventually recover. Fourth, decisions about continuing treatment and life support should be individualised, guided by multidisciplinary teams in collaboration with families and ethical consultants. These decisions must consider the potential for recovery as highlighted by this study, while acknowledging the absence of definitive prognostic biomarkers.

Although the mechanisms underlying prolonged impaired consciousness are likely multifactorial and remain unclear, persistently high CSF antibody levels in late disease stages likely sustain synaptic and circuit dysfunction. Future studies should integrate biomarkers beyond antibody titres (eg, cytokines and chemokines, complement, neurofilament light chain, Fc glycosylation[30]) with standardised neurobehavioural assessments, and recommended electrophysiological and neuroimaging evaluations.[11,18] Evidence highlights the urgent need for NMDAR encephalitis-specific trials to assess novel treatment strategies, with protocols adapted to the complexities of evaluating consciousness and treatment response.

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**Contributors**

MG participated in the study design and conceptualisation, data collection, data interpretation, statistical analysis, figure, table, and video creation, antibody studies, literature search and review, and writing and critical approval of the paper, and obtained funding. MMS participated in data collection, data interpretation, figure and video creation, and critical approval of the paper. GM-S, JB, KWa, KWu, PHL, GO-C, JY, KPW, SM, GDL, EL, TC, MJM-G, JPF-C, MSh, EF, TM, HN, TT, SS, YI, SN, AH, MT, AI, DW, MSp, YB, RDP, KWS, FL, RH, CF, MJT, and TA participated in data collection, data interpretation, and critical approval of the final paper. RB participated in statistical analysis and critical approval of the final paper. EA and LM participated in antibody studies and critical approval of the final paper. JP participated in figure creation and critical approval of the final paper. FG participated in the study conceptualisation, data interpretation, and writing and critical approval of the paper. TI participated in data collection, data interpretation, figure and video creation, and critical approval of the paper. JD participated in the study design and conceptualisation, data interpretation, figure, table, and video creation, literature review, and writing and critical approval of the paper, and obtained funding. MG and JD accessed and verified the data. All authors were permitted full access to all the data in the study and had final responsibility for the decision to submit for publication.

**Declaration of interests**

JD holds patents licensed and receives royalties to Euroimmun for the use of NMDA, GABAB receptor, GABAA receptor, DPPX, and IgLON5 as autoantibody tests, for which he receives royalties. FG holds a patent licensed to Euroimmun for the use of IgLON5 in an autoantibody test, for which he receives royalties. He receives honoraria from MedLink Neurology for his role as associate editor. JB and MJT hold a copyright, on behalf of Erasmus MC, for the patient-reported outcome measure PROSE. JB is supported by the AEA Seed Grant 2024. The Medical University of Vienna, Austria (employer of RH) receives payment for antibody assays and for antibody validation experiments organised by Euroimmun (Lübeck, Germany). All other authors declare no competing interests.

**Data sharing**

Anonymised participant data can be shared with qualified investigators on request to the corresponding authors, from the date of publication onwards.

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